

Linear Attention: buying a memory that stops growing, and paying for it in recall

One-page concept summary · DataForge 2026, Pathway Track · companion to the interactive explainer *Where Memory Breaks*
For a data scientist meeting linear attention for the first time. Prerequisites: dot products, matrix–vector multiplication.

The design pressure

A Transformer scores each query against every earlier token, so it must retain every key and value it has seen. That KV cache grows with context length and bottlenecks inference memory [1]. Linear attention removes the all-pairs comparison by reordering the computation: instead of comparing token pairs, it carries one fixed-size state that each token updates additively, $S \leftarrow S + \varphi(k_i) \otimes v_i$ and answers a query by reading $r(q) = \varphi(q)^T S$. Per-token cost becomes constant in sequence length, and the state has the same shape after ten tokens as after ten million.

What the change costs

The state is a superposition: every association is summed into the same matrix, so nothing is stored separately. Probing it with the key that wrote a fact returns that fact *plus* a weighted sum of every other stored value:

$$r(k_i) = (k_i^T k_i) v_i + \sum_{j \neq i} (k_i^T k_j) v_j$$

The central claim our explainer teaches: a fixed-shape state can absorb an unbounded stream of key–value facts, but because all writes are superposed, retrieval degrades as facts accumulate or as keys grow more similar — and enlarging the state postpones that failure rather than removing it.

Arora et al. identify this as a direct state-size/recall trade-off and show that fixed-state alternatives such as H3, Mamba and RWKV "struggle at recall"; their BASED architecture makes state size a dial traversing the recall–memory Pareto frontier, beating comparable sub-quadratic models on recall-intensive tasks by 10.36 accuracy points [1]. Pandey & Singh sharpen the cause: the state grows as $O(T)$ in Frobenius norm, "causing progressive interference between stored associations" [2]. Zhong et al. formalise the same separation as a retrieval signal-to-noise ratio under an associative-memory reading of attention [3].

Three responses, and what each gives up

Approach	Memory rule	Recall behaviour	Evidence level
Softmax attention	Keeps every key–value pair; grows with context.	Exact — the reference. Cost, not accuracy, is the problem.	Ubiquitous production use.
Linear attention (kernel / normalized)	One fixed state, additive writes; feature map and denominator control scale.	Degrades once stored pairs exceed state capacity; normalization controls magnitude, not interference.	Author-reported benchmarks, incl. MQAR [1].
Gated DeltaNet	Fixed state; gating erases while the delta rule targets updates, so a write can edit rather than merely add.	Surpasses Mamba2 and DeltaNet on in-context retrieval, length extrapolation, long-context understanding.	Author benchmarks, ICLR 2025 [4]; separately reported in Qwen3-Next — deployment signal, not independent evaluation.
Variational linear attention	Fixed state; update recast as regularised online least squares with a Sherman–Morrison rank-1 penalty.	Cuts $\ S_t\ _F$ 109× at $T=1,000$; near-exact MQAR while pairs < head dimension; 62% at that boundary.	Proofs plus author-run experiments, 2026 preprint; no independent replication yet [2].
BDH-CQ contextual memory	Recurrent state updated by a <i>learned</i> function, once per demonstration rather than per token.	Adaptation happens in recurrent state, not weights — no parameter updates at inference.	Developer-reported.

Where BDH and BDH-CQ actually fit

Their relevance is real but unequal. **BDH** (Dragon Hatchling) reformulates attention as synaptic memory: reasoning and memory share one computational fabric, associations are written by Hebbian updates between co-active neurons, and activity is sparse and non-negative — roughly 5% of neurons active in reported runs. Its GPU-oriented formulation, BDH-GPU, is built from ReLU-low-rank transformations *with* linear attention, which is the direct link to this concept; it is explicitly not a state-space model in the Mamba sense. **BDH-CQ** is the closer case: its contextual memory is published as a general learned update $S_t \leftarrow U_\theta(S_{t-1}, D_i)$, and its report relates that memory to attention, fast-weight memory and linear-attention views of contextual association through an *additive special case* where state accumulates additively per demonstration. That special case — and only that — is what our explainer reimplements, as a labelled synthetic toy. We do not run BDH-CQ weights, and U_θ is not assumed additive.

The most important open question

Not whether gating or regularisation helps — both demonstrably do — but whether *any* fixed-size state can hold what a long-context task requires without growing the state or smuggling exact storage back in. Both leading fixes concede the capacity boundary rather than dissolve it: VLA's accuracy holds while stored pairs stay below the per-head dimension and drops to 62% at that boundary [2], and BASED's answer is explicitly a hybrid — linear attention *plus* sliding-window attention, state size as a knob [1]. Independent reproduction is the missing evidence throughout: the 2026 variational results and the BDH/BDH-CQ results are, as of writing, reported by their own authors.

Continue here

- Arora, Eyuboglu, Zhang, et al. *Simple linear attention language models balance the recall-throughput tradeoff* (BASED). arXiv:2402.18668, 2024.
- Pandey & Singh. *Variational Linear Attention: Stable Associative Memory for Long-Context Transformers*. arXiv:2605.11196, 2026.
- Zhong, Xu, Ao & Shi (ByteDance Seed). *Understanding Transformer from the Perspective of Associative Memory*. arXiv:2505.19488, 2025.
- Yang, Kautz & Hatamizadeh. *Gated Delta Networks: Improving Mamba2 with Delta Rule*. arXiv:2412.06464, ICLR 2025.
- Arora et al. *Zoology: Measuring and Improving Recall in Efficient Language Models*. arXiv:2312.04927, ICLR 2024 — origin of the MQAR task our explainer runs live.
- Pathway. *The Dragon Hatchling and the BDH-CQ technical report* — primary sources for the BDH claims.

The companion artifact lets a reader set the number of stored facts and the state size and watch retrieval accuracy, the state's Frobenius norm and the recovered values update live — the claim above is falsifiable in under a minute. Live computation runs on synthetic data (10-value vocabulary, random Gaussian keys, no learned projections); every illustrative or precomputed element is labelled as such in the artifact and its README.